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Exploring the mammalian sensory space: co-operations and trade-offs among senses

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Abstract The evolution of a particular sensory organ is often discussed with no consideration of the roles played by other senses. Here, we treat mammalian vision, olfaction and hearing as an interconnected whole, a three-dimensional sensory space, evolving in response to ecological challenges. Until now, there has been no quantitative method for estimating how much a particular animal invests in its different senses. We propose an anatomical measure based on sensory organ sizes. Dimensions of functional importance are defined and measured, and normalized in relation to animal mass. For 119 taxonomically and ecologically diverse species, we can define the position of the species in a three-dimensional sensory space. Thus, we can ask questions related to possible trade-off vs. co-operation among senses. More generally, our method allows morphologists to identify sensory organ combinations that are characteristic of particular ecological niches. After normalization for animal size, we note that arboreal mammals tend to have larger eyes and smaller noses than terrestrial mammals. On the other hand, we observe a strong correlation between eyes and ears, indicating that co-operation between vision and hearing is a general mammalian feature. For some groups of mammals we note a correlation, and possible co-operation between olfaction and whiskers.

Keywords Sensory ecology · Vision · Olfaction · Hearing · Vibrissae

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Introduction

Many textbooks and review papers give a broad presentation of mammalian senses (Barlow and Mollon 1982; Plassmann and Brändle 1992; Smith 2000; Supin et al. 2001; Peichl 2005), and some focus on behavioral ecology and evolution with examples of co-operation among senses (Dusenbery 1992; Heffner and Heffner 1992; Bradbury and Vehrencamp 1998; Barth et al. 2012; Stevens 2013). However, to our knowledge, no study has tried to create a quantitative system for evaluation of the co-operation vs. trade-off between different sensory systems. This paper is an attempt in that direction. We use sensory organ size for estimating how much an animal invests in a particular sensory system. Our material covers many order-level clades and ecological niches. However, we have excluded whales, as their olfactory organs (Oelschläger 1989) and ears (Fleischer 1978; Ketten 2000; Nummela et al. 2004; Hemilä et al. 2010) radically differ from those of terrestrial mammals.

Using organ sizes, we thus estimate the relative roles of vision, olfaction and hearing (eyes, noses and ears) in the life histories of 119 mammalian species. For a smaller group of species we include a fourth dimension, i.e., whiskers and other types of tactile nose receptors. A major

underlying assumption, behind the use of organ size as indicator, is that sensory organs are metabolically expensive and that an investment in a large organ is an indirect indication of its functional importance (for published evidence, see "Discussion").

How does sensory organ size affect perception?

Another underlying assumption is, of course, that larger organs may be more efficient. Here, we investigate the situation for eyes, noses and ears (schematically presented in Fig. 1, left column).

Eyes

Vision has many dimensions. Here, we describe the anatomical basis of visual acuity during daylight vision and the general photosensitivity of night-time vision. The former is cone-based photopic vision, and the latter rod-based scotopic vision. Besides some photopic primates and some

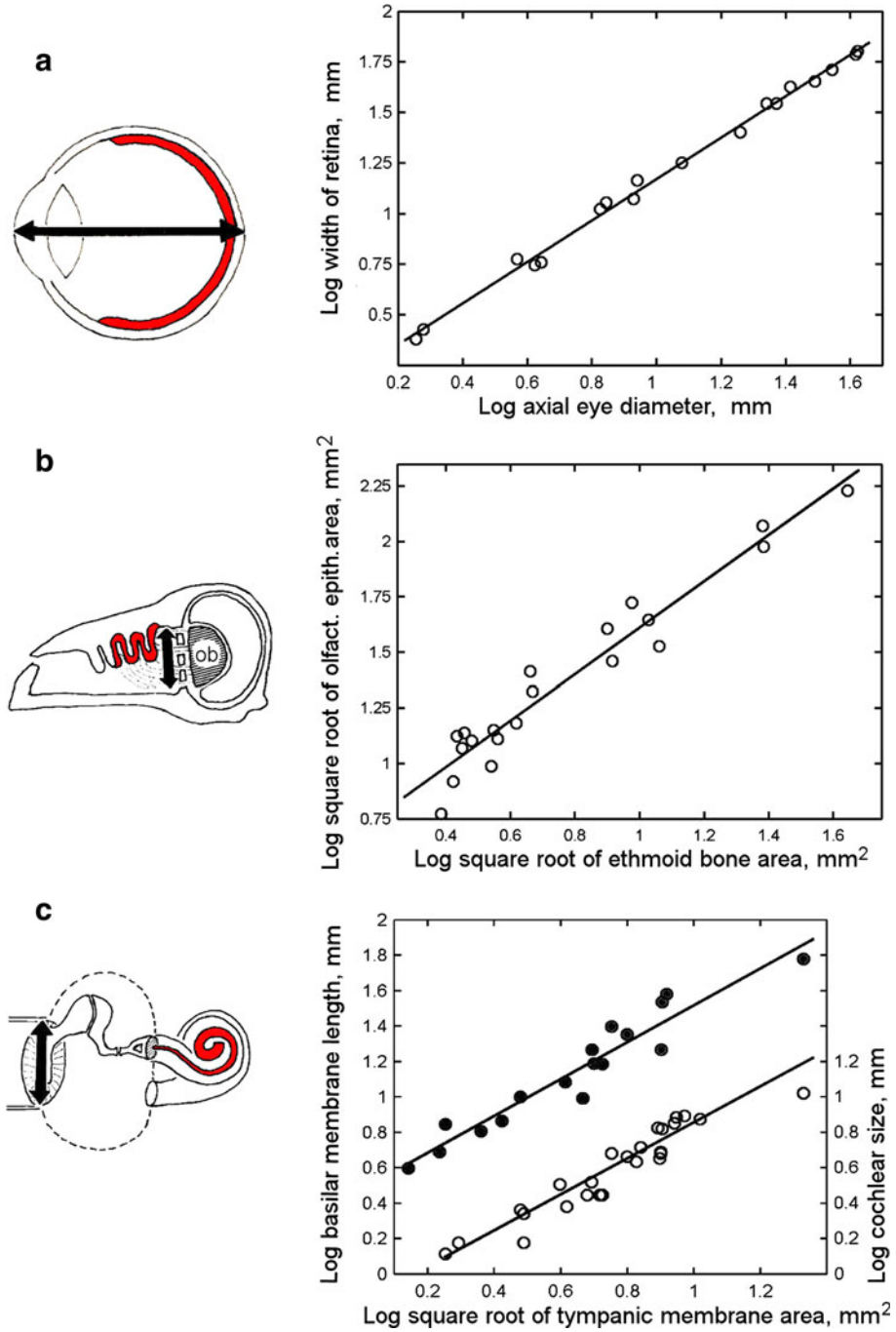


Fig. 1 The isometry of three sensory organs illustrated by plotting log sizes of the sensory cell epithelia as a function of log sizes of respective sensory organ. *Schematic drawings* show sensory epithelia (*red*) and structures used for measuring organ size (*black arrows*). **a** Log nasal–caudal (or nasal–temporal) diameter ("width") of spread-out retinas, as function of log axial eye diameter; **b** log square root of olfactory epithelium area, as function of log square root of cribriform plate area; "ob" = olfactory bulb; **c** *black circles*, left *y* axis: log basilar membrane length as function of log square root of tympanic membrane area; *open circles*, *right y* axis: log cochlear size (as defined by Spoor et al. 2002), as function of log square root of tympanic membrane area. The cochlear size/tympanic membrane relation of microchiropteran bats and otariid seals deviates from the typical mammalian pattern; thus, two bats (*Rhinolophus*, *Pipistrellus*) and one otariid seal (*Zalophus*) are excluded from Fig. 1 and from the isometry calculations. They are, however, included in Supplementary Material Table S1. The 17–25 species used for the plots represent those mammals for which data for the size of sensory epithelia exist, and they do not in all cases overlap with the 119 species composing our main data set. The *straight lines* are least-square regression lines; see Text for the respective coefficients

exclusively scotopic mammals, there is a large group characterized as cathemeral, i.e. both day- and night-active animals. Visual acuity in twilight and daylight is a function of two anatomical parameters: the size of the eye, and the density of cones and ganglion cells in the retinal region responsible for high-acuity vision, i.e., for most mammals an area centralis, and for many primates a fovea (Hughes 1977; Barlow 1981; Henderson 1985; Hirsch and Curcio 1989; Stone and Halasz 1989; Peichl 1992; Kiltie 2000; Mass and Supin 2003; Reuter and Peichl 2008).

Mammals direct their vision, and retinal areas of high density of visual cells, toward objects of interest. Thus, images of these objects are automatically projected onto the most sensitive part of the retina. The size of the projection is proportional to the size of the eye, and within a group of mammals with similar cone and ganglion cell densities, those with larger eyes have a higher visual acuity, much like a large photograph can reproduce more detail than a small one (assuming the same pixel size in both of them). An approximate visual acuity can be calculated from the posterior nodal distance of the eye (PND), and the maximum ganglion cell density within the area centralis (Pettigrew et al. 1988; Mass and Supin 1992, 2000; Ahnelt and Kolb 2000; Supin et al. 2001). Such estimations agree reasonably well with acuities observed in behavioral studies, and a calculation carried out for eight terrestrial mammals, all of them cathemeral, show the expected relation between calculated acuity and eye size (Reuter and Peichl 2008, Table 10.2 and Fig. 10.6; Pettigrew et al. 2010). Thus, for cathemeral mammals, visual acuity increases in a regular way with increasing eye size.

Absolute scotopic sensitivity to very weak stimuli in full darkness is a special case where threshold limits are set by the rhodopsin absorbance in the retina and the total flux of photons entering the pupil. In cathemeral and scotopic mammals the maximum size of the pupil is a function of eye size. Thus, increasing eye size tends to favor also absolute sensitivity at night.

Noses

Studying olfactory thresholds in behavioral experiments is technically difficult, although descriptive anatomical works (Mulvaney and Heist 1970; Kolb 1971, 1975; Güntherschulze 1979; Söllner and Kraft 1980) as well as quantitative works do exist (Bhatnagar and Kallen 1975; Gross et al. 1982; Sigmund and Sedláček 1985; Rowe et al. 2005). Pihlström et al. (2005) gathered data for eight mammalian species showing a continuous increase in sensitivity to butyric acid (a metabolite excreted by most mammals) when advancing from a small bat to the German shepherd. Compared with the bat, the latter has a 70 times larger olfactory organ. Man and the hedgehog were found to have similar "middle-sized" olfactory organs and similar sensitivities to many odors, among them butyric acid.

Data about olfactory epithelium areas are also scarce, but as epithelium area is proportional to the area of the perforated ethmoid bone (Fig. 1b), we predict that not only the German shepherd but also the wild boar and the Asian elephant, and many other ungulates with large ethmoid bones, have sensitive olfaction, a prediction being in accordance with qualitative observations (Pihlström et al. 2005). Other cranial features related to sensory systems of the head have suggested that invasion of an aquatic environment results in the reduction of olfaction, and hence in reduced sensitivity when compared to terrestrial taxa (Sánchez-Villagra and Asher 2002).

Ears

The size of the middle ear and the cochlea (Fig. 1c, left column) is correlated with the frequency range to which the animal is evolutionarily adapted (Bruns et al. 1983–1984; Pye 1985–86; Müller et al. 1993; Lange et al. 2004; Nummela and Sánchez-Villagra 2006). Elephants and humans have large ears and massive middle ear ossicles, and their sensitive hearing is restricted to frequencies below 10 and 20 kHz, respectively, (Heffner and Heffner 1980; Evans 1982). On the other hand, mice and bats with their small ears hear reasonably well up to 100 and 120 kHz, respectively, (Henson 1961; Dalland 1965; Heffner and Masterton 1980; Fay 1988; Heffner et al. 2001; Heffner and Heffner 2010). The fact that hearing of large mammals is restricted to low frequencies is at least partly due to the inertia resulting from large ossicle mass (Hemilä et al. 1995, Fig. 4). However, the size of the ossicles does not significantly affect thresholds in the frequency region of optimum sensitivity.

There are some aspects of hearing where a larger ear seems to increase the amount of information received. Most important may not be the absolute sensitivity of hearing, but the ability to detect minor changes in sound level and sound frequency, and there a large ear with a long basilar membrane may be of crucial importance. Following up the classical studies of Békésy (1960), Fay (1988, 1992) collected psychophysical data about frequency discrimination in ten mammalian species, and related these data to basilar membrane lengths. The scatter of data points is large, but the conclusion is clear: the number of independent frequency information channels increases with increasing absolute basilar membrane length (Fay 1992, Table 14.1, Fig. 14.13).

Two main mechanisms for frequency discrimination have been proposed. Békésy (1960) and Greenwood (1962, 1990) combined biophysical information about the "traveling

wave" passing along the basilar membrane in the cochlea, with psychophysical data about human and animal frequency discrimination. They observed that different sound frequencies produce traveling wave maxima at different positions along the basilar membrane, and suggested that frequency information is transmitted via ganglion cells positioned at these maximally vibrating membrane regions.

Other authors suggest that frequency is partly encoded by phase-locked firing rates in these ganglion cells (phase locking means that action potentials are at least partly synchronized to stimulus waveform, Geisler 1998, p. 197). These two mechanisms may complement each other, and both may be especially efficient in large ears. Firstly large ears have long basilar membranes (Fig. 1c), and secondly efficient phase locking is possible only at sound frequencies below about 3 or 4 kHz (Louage et al. 2005), i.e., at frequencies being outside the main hearing range of very small mammals (Fay 1988; Heffner et al. 2001, Fig. 4). The overlap between hearing range and phase-locking region may thus gradually increase with increasing ear size.

When comparing hearing ranges of different mammals, behavioral thresholds are usually plotted on logarithmic scales. In this format, most mid-sized and large mammals have audiograms of similar shape, although differently positioned on the log frequency scale (Nummela 1997). However, within a range of smaller species, from mice to domestic cats, the larger species with larger ears cover a wider range of log frequencies (West 1985; Fay 1988; Hemilä et al. 1995, Fig. 8; Heffner et al. 2001; Manoussaki et al. 2008). This can be seen as a broadening of the acoustic information received when going from small mice to rabbits and cats, and we hypothesize that, in this size region, the role of phase-locking may increase with increasing animal size. Hence, there are a number of mechanisms indicating that large ears favor high-quality hearing.

Isometry

A condition for letting a single number represent the size of a complex organ such as the ear or the olfactory organ (nose), is that the organ (or relevant parts of it) has an approximately isometric structure, i.e., a similar shape independent of absolute size, or scale (Kendall 1989). The mammalian eye is a roughly spherical optical system that projects images onto the light-sensitive and image-analyzing retina at the back of the eye (Rodieck 1998). In other words, in this organ structure and function are tightly knit and the eye of a terrestrial mammal can be considered a model for a perfectly isometric sensory organ. Approximate isometry has also been observed for the mammalian middle ear (Rosowski and Graybeal 1991; Nummela 1995; Hemilä et al. 1995; Nummela and Sánchez-Villagra 2006), and for the area of the cribriform plate of the ethmoid, i.e., the perforated bony plate through which the axons of the olfactory receptor cells pass before they reach the olfactory bulb (Fig. 1b: ob) (Pihlström et al. 2005 and this study). Thus, these organs, or corresponding proxy measures, are basically suitable for comparisons based on size.

Materials and methods

Sensory organ data from skull structures and soft eye tissues

The primary data consist of four measures of the 119 species from which we have obtained complete information: body mass (mainly from Nowak 1999), and the sizes of eyes, olfactory organs and ears (Supplementary Material Table S2). In the same Table, we present information on typical habitats and diets for all species included. To avoid bias in the selection of data, we have drawn this information from a single source (Nowak 1999). We delimit the sensory organs in the following way: "eye" means only the eyeball; "olfactory organ" includes the nasal cavity with the olfactory epithelium, the cribriform plate and the olfactory bulb, but not various peripheral nasal structures; "ear" includes the middle ear from the tympanic membrane to the oval window, and the cochlea, but not the auditory canal nor the highly variable outer ear pinnae (Fig. 1, drawings). The reason for these limitations is that the structural uniformity of sensory organs applies to some functionally basic structures but does not apply to outer ears and the very variable morphology of mammalian noses and trunks. The distal parts of these sensory organs, being in direct contact with a variable outer world, are correspondingly diverse.

Measurements were made on hundreds of skulls. Complete sets of data for all three sensory organs were obtained from only 119 species. These include 33 carnivores, 18 primates, 16 rodents, 13 artiodactyls, 8 chiropterans, and 31 species representing other orders. The material covers therian mammals rather well, including placentals and marsupials, but has a bias toward large species.

Eye data were obtained from published sources describing soft eye tissues in adult mammals (i.a., Steindorff 1947; Stone 1965; Hughes 1975; Ritland 1982; Gonzalez-Soriano et al. 1997; Howland et al. 2004). Olfactory organ and middle ear sizes were determined from skull material. For determining the area of the perforated cribriform plate, we used endocranial casts; heated and fluid 'vinamold' was poured through the foramen magnum into the foremost part of the skull where it fills the cavities originally surrounding the olfactory bulb. After cooling, this solid but elastic cast was detached and

removed through the foramen magnum. Even when stretched slightly, this material retains its shape, and hence its dimensions can be measured (Pihlström et al. 2005). Using suitable digital calipers, the area of the tympanic membrane was measured directly from the sulcus tympanicus, the bony ridge surrounding the membrane (Nummela 1995). Usually one species is represented by 1–3 adult animals; and when several specimens are measured, a mean is calculated. Information about sex is missing for many museum specimens and is therefore not given here.

Taxonomy and nomenclature follow Wilson and Reeder (2005). The material studied is deposited at the Finnish Museum of Natural History in Helsinki, the Swedish Museum of Natural History in Stockholm, the Zoological Museum in Copenhagen, Denmark, and the Senckenberg Research Institute, Frankfurt.

Numerical analysis

As the absolute size of organs correlates strongly with animal body size, many interesting comparisons especially regarding relative investments in different senses require that this dominant variable be appropriately factored out. This was done as follows.

Body size was measured as body mass in kilograms (basically proportional to the third power of the animal's linear dimensions). This was used as the independent variable denoted Mass. The sensory variables representing organ size were all first transformed to linear dimensionality. These dependent variables are the axial length of the eye in mm, termed *Eye*, the square root of the cribriform plate area in mm^2, termed *Nose*, and the square root of the tympanic membrane area in mm^2, termed *Ear*. When the (base 10) logarithm of each of these as function of log Mass (see Fig. 3) is fitted by least-square linear regression, the following "ordinary least squares (OLS) models" are obtained for the three sensory variables:

$$\log \text{Eye} = 0.937 + 0.244 \log \text{Mass} \quad (1)$$

$$\log \text{Nose} = 0.856 + 0.225 \log \text{Mass} \quad (2)$$

$$\log \text{Ear} = 0.610 + 0.145 \log \text{Mass}. \quad (3)$$

All of the six correlation coefficients in Eqs. (1–3) are statistically significant ($p < 0.001$, Student's t test; Lehmann and Romano 2005).We tested least-square fitting also of second-order polynomials, i.e., including an additional quadratic mass term $(\log \text{Mass})^2$. We found that the coefficients of the quadratic terms are not significantly different from zero for nose and ear ($p > 0.01$, Student's t test), whereas the coefficient of the quadratic term in the log Eye vs. log Mass function is indeed significantly different from zero ($p < 0.001$, Student's t test):

$$\log \text{Eye} = 0.625 + 0.157 \log \text{Mass} - 0.010 (\log \text{Mass})^2. \quad (4)$$

The meaning of the negative quadratic term in Eq. (4) is seen by inspecting Fig. 3, where the dependence of log Eye on log Mass is seen to be somewhat concave. The interpretation of curvilinearity is discussed later in the section "Filtering out the scale effect". In the plots and numerical analysis of residual organ sizes, however, we use the "OLS models" based on linear regression on the log–log data, given by Eqs. (1–3). Briefly, the metrics of the *body mass-independent sensory space* is defined by the residuals, i.e., the differences between the measured variables and the values given by Eqs. (1–3). In other words, the coordinates of each species in the three-dimensional sensory space are the residuals of log eye, log nose and log ear.

The performance of the models can be described by the coefficient of determination r^2, which expresses the fraction of the total variation of the dependent variable that is explained by the model. For the OLS model used here, it is equivalent to Pearson's coefficient of linear correlation squared. For the three Eqs. (1–3), $r^2 = 0.756$ (log eye), 0.733 (log nose) and 0.763 (log ear), respectively, (cf. the linear correlation coefficients given in Fig. 2).

Results

Sensory organ isometry

We have investigated the isometry of mammalian eyes, noses, and ears by plotting the size of the epithelia carrying the sensory receptor cells (i.e., the retina, the olfactory epithelium, and the basilar membrane of the cochlea, respectively) as a function of the size of the structure used for measuring the respective organ size. The measures used for the sizes of eyes, noses and ears are: (1) the axial diameter of the eye, (2) the area of the cribriform plate, i.e., the perforated ethmoid bone plate at the caudal end of the nasal cavity, and (3) the area of the tympanic membrane (Fig. 1; Supplementary Material Table S1). Area measures were converted into linear measures by taking their square root. The linear variables were plotted on logarithmic scales, and fitted with least-square regression lines. The plots indicate almost perfect isometry, i.e., the regression coefficient k is in all cases close to one. The high r^2 values imply a modest scatter of data points in relation to the regression lines. Thus, the condition for letting a single linear value represent linear organ size is fulfilled.

The parameters k and c of the regression lines $y = kx + c$, and the coefficients of determination r^2, are as follows:

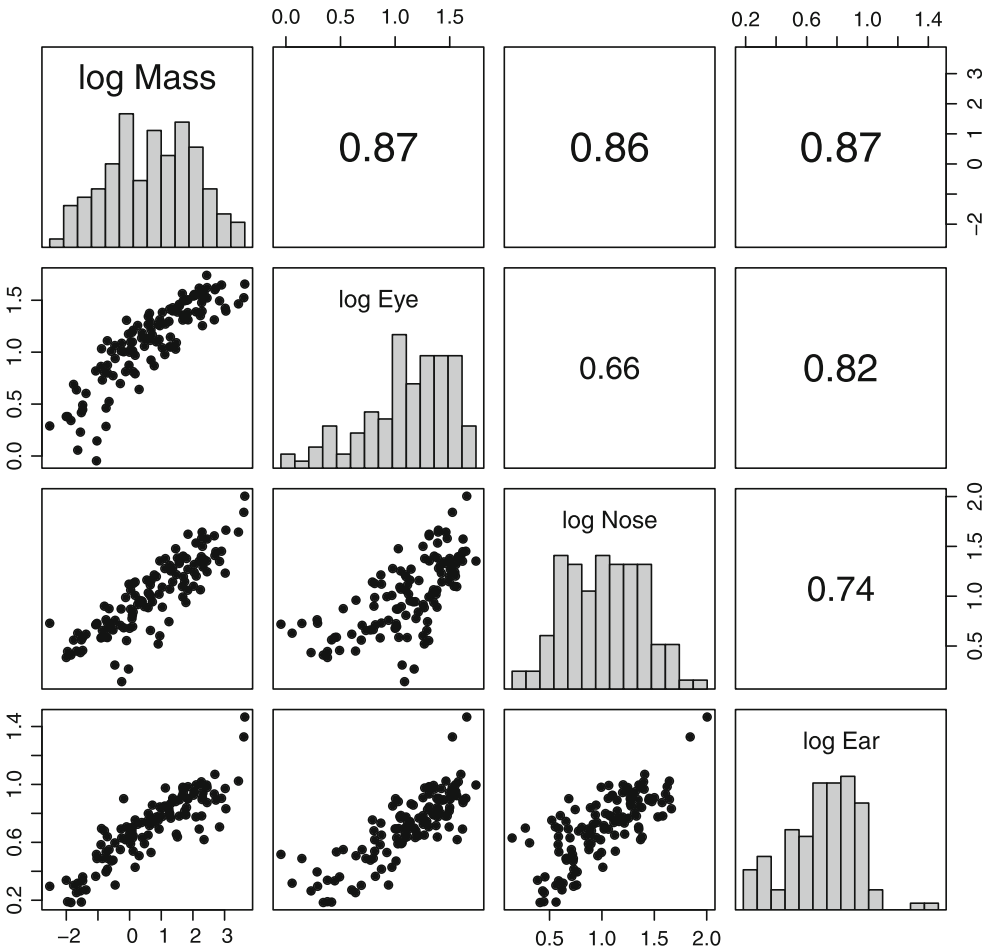


Fig. 2 *Scatter plot* matrix presenting the mutual relations between four size parameters, (*1*) log mass (kg), (*2*) log eye (mm), (*3*) log nose (mm), and (*4*) log ear (mm) (see Text for the definition of these). The three blocks in the leftmost column show log organ size (*y* axis) as function of log body mass (*x* axis). Each *small circle* represents one species. The remaining *three blocks* show eye size as function of nose size etc., i.e., the mutual relations between log organ sizes. *Gray bar charts* at the *diagonal* show relative number of species and their size distribution within each category, i.e., categories of log mass, log eye, log nose, and log ear. Numbers in the *upper right squares* give Pearson's coefficients of linear correlation between the respective pairs of variables. These correlations are never very small, as all variables contain a component related to body mass

Fig. 1a $k = 1.023$, $c = 0.147$, $r^2 = 0.995$
Fig. 1b $k = 1.046$, $c = 0.565$, $r^2 = 0.940$
Fig. 1c (above) $k = 1.041$, $c = 0.476$, $r^2 = 0.917$
Fig. 1c (below) $k = 0.983$, $c = -0.153$, $r^2 = 0.898$

In Supplementary Material Table S1 two echolocating bats and one otariid seal are enclosed within brackets, and they are not included in the above analysis in Fig. 1. They are clear outliers with large cochleae in relation to their small tympanic membranes. In the former case, we note that hearing high-frequency echoes requires small and stiff tympanic membranes (Rosowski 1992), and extracting echo information may require large cochleae (Pollak 1992).

Correlations and co-operations among senses

The mutual relations between animal mass (kg), and sensory organ sizes (mm), are presented in a scatter plot matrix (Fig. 2). The upper right part of the matrix shows correlation coefficients calculated from the data in six blocks presented in the lower left part of the Figure. All blocks contain data for all 119 species. Three blocks in the leftmost column show a strong positive correlation for log body mass in relation to the logarithms of all three sensory organs (coefficients 0.86–0.87). The three blocks in the middle part show the mutual relations between the three sensory organs: log nose size as a function of eye size (correlation coefficient 0.66), log ear size as a function of eye size (0.82), and log ear size as a function of nose (0.74). None of the coefficients is very low as all variables contain a component related to body size. The coefficient 0.66 for the nose–eye pair is statistically different from the coefficient 0.82 of the ear–eye pair.

The high correlation between ear and eye is in accordance with the dense cluster of data points in the ear–eye block, clearly denser than the corresponding swarm of dots in the nose–eye block. In our interpretation, this indicates an anatomical evolution with co-operation between vision and hearing. Animals with large eyes (in relation to body size) tend to have large ears too. This tendency is illuminated by a rare exception; the two elephants, *Elephas* and *Loxodonta*, have very large tympanic membranes (ears), but rather small eyes (in relation to body mass). In the ear–eye block they are seen as two dots in the upper right

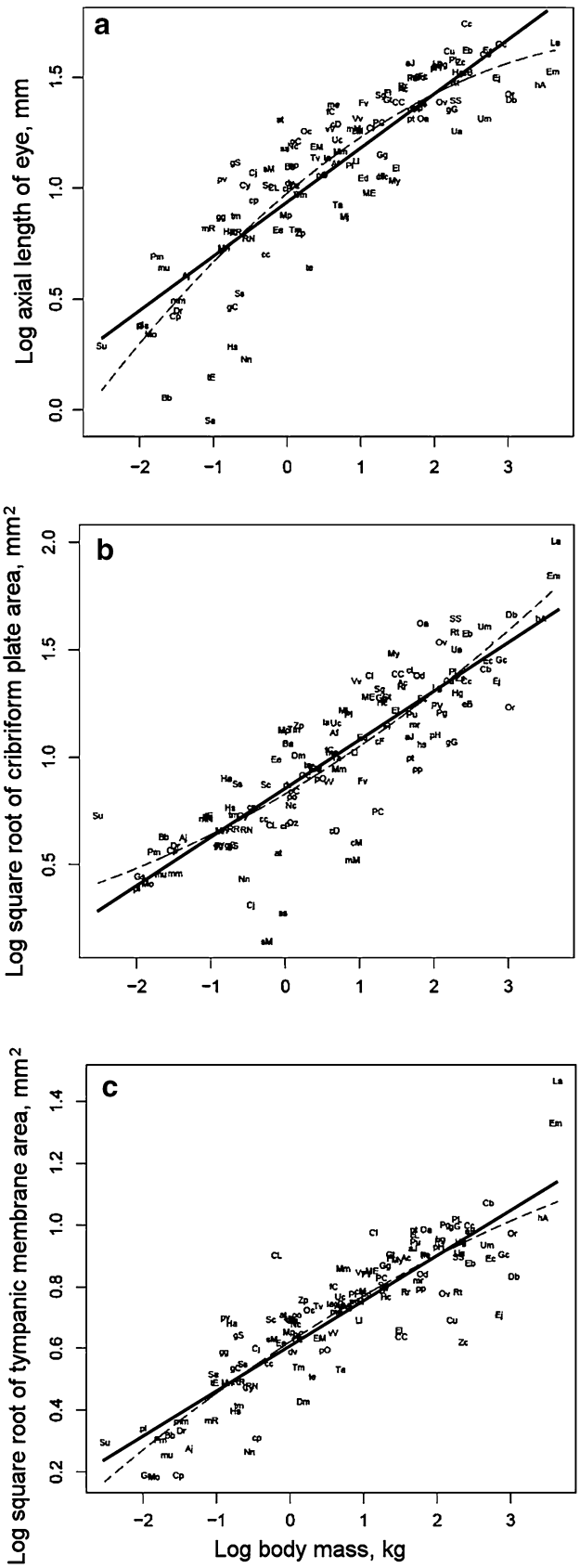


Fig. 3 Linear measures of sensory organ size (mm) as function of animal mass (kg) plotted on log–log axes for all three sensory organs. A two-letter code (see Table 1 and Supplementary Material Table S2) gives the approximate position of each species. The *straight lines* represent least-square linear regression fits (the OLS model), *dashed curves* quadratic fits (see "Numerical analysis" in "Materials and methods")

corner, high above the cluster representing the typical situation. Thus, our conclusion is that the high correlation coefficient for the ear–eye pair, clearly higher than that of the nose–ear pair, indicates that a specific co-operation between vision and hearing is a general mammalian feature, not only an interesting characteristic of primates and horses (see "Discussion").

Filtering out scale effect

The filtering out of the scale effect was carried out as follows. The regression lines in Fig. 3a–c describe estimated sizes of sensory organs and they represent "typical" organ sizes with a residual of zero, while larger organs are given positive and smaller organs negative residual values (see "Materials and methods", "Numerical analysis"). Eye diameters saturate for large mammals, e.g., elephants do not have larger eyes than horses and large pinnipeds. A similar saturating tendency may be visible for ears (although elephant tympanic membranes are uniquely large), while the relation between log cribriform plate size and log animal mass may be described by a single allometric relation across the whole range of body sizes. The spread of data points above and below the line is large. The deviation of each point from the line is the residual organ size, where the general effect of body size has been filtered out (Supplementary Material Table S2). The residual size thus gives log organ size in relation to the typical size observed in mammals having the same body mass. Using residual sensory variables, i.e., animal size-corrected values, we may also compare data from small and large animals.

As described in the section "Numerical analysis" in "Materials and methods", the filtering out of the scale effect relies on the simple model marked by straight lines in the log–log plots of Fig. 3a–c, implying that the relation between organ size and body size follows a single allometric relation over the whole size range. This simplifies calculations and facilitates application of the method. We have, however, also carried out calculations based on least-square fittings using an additional quadratic mass term (see above, "Numerical analysis"). For the smallest and the largest species this causes some changes, which do not, however, produce statistically significant changes for noses and ears, and do not affect our main results and conclusions.

The sensory space concept

The term sensory space used in this paper does not, of course, refer to a physical space; it is a metaphor suggesting that we can, in a way analogous to the mapping of the extension of an object within a three-dimensional physical space (with axes *x*, *y*, and *z*), describe the sensory profile of an animal by referring to the sizes of three sensory organs. Sensory space may thus be considered as trait-based ecometrics sensu Polly et al. (2011), scalable across hierarchies and scales. As explained in the "Introduction" section, we have reason to believe that the relative size of an eye or nose correlates with sensitivity and resolution. We also pointed out that for ears, the relation between size and performance is more complex.

To illustrate the concept, we first present the species positions in sensory space based on residual organ sizes of 21 species out of the 119 in the complete material. They represent 11 carnivores and 10 primates (Fig. 4). Four of the carnivores are canids ("dogs", red) and seven are felids ("cats", orange), while six of the primates are strepsirrhines ("prosimians", blue), and four are hominids ("apes", green). These groups are complete in the sense that, among the 119 species that were investigated, those presented in the diagram represent all studied canids, felids, strepsirrhines, and hominids.

The black dot in the center of the space in Fig. 4 is the common zero point for all three axes: the horizontal *x* axis (eye axis) is to the right, the horizontal *y* axis (nose axis) to the left, and the vertical *z* axis represents ear size. The numerical graduations along the outer borders of the space mark log residual organ sizes (data from Supplementary Material Table S2). The black zero dot in the center marks, for all three senses, normal organ size (i.e., the lines in Fig. 3a–c). Filtering out the scale effect focuses the three lines to a common point, and all data points above the lines in Fig. 3a–c are now on the plus side, whereas all points below are on the minus side of the axes.

Each colored dot in Fig. 4 defines the position of a given animal species in relation to all three axes in this residual space; thus, the Figure is based on $3 \times 21 = 63$ residual values. The end points of the vertical sticks mark the positions of the 21 species onto the horizontal plane indicating the zero level for ear size. The corresponding zero levels for eye and nose form two vertical planes at right angles in relation to each other and to the horizontal ear plane. Only three species have clearly negative ear values, while three others lie close to the zero level.

The data in Fig. 4 demonstrate the discriminating power of the method. Canids and felids are well separated, as are the two groups of primates. Among carnivores, canids differ from felids by having larger olfactory organs, and among primates, prosimians differ from apes by having larger eyes. The *z* axis, marking ear size, does not discriminate between the four groups presented here.

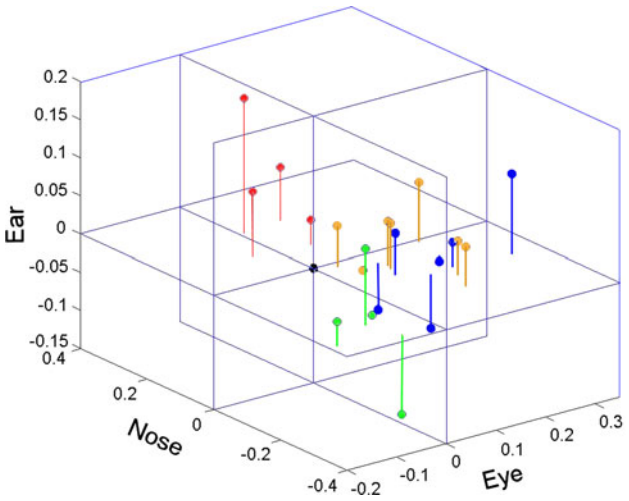


Fig. 4 Diagram showing the positions of 21 mammalian species in a residual three-dimensional sensory space. *Vertical z* axis represents residual log ear size, *horizontal y* axis residual log nose size (*left scale*), and *horizontal x* axis residual log eye size (*right scale*). The *black dot* in the center is the origin of all three axes. Four groups are distinguished by different *colors*: "dogs" (*red*), "cats" (*orange*), prosimians (*blue*), and apes (*green*). The end points of the *vertical sticks mark* the positions of the 21 species projected onto the *horizontal* plane for zero ear. The four "dogs" are *Canis latrans*, *C. lupus*, *Urocyon cinereoargenteus*, and *Vulpes vulpes*; the seven "cats" are *Acinonyx jubatus*, *Felis catus*, *Lynx lynx*, *Panthera leo*, *P. uncia, Prionailurus viverrinus*, and *Puma concolor*; the six "prosimians" are *Galago senegalensis*, *Eulemur macaco*, *Nycticebus coucang*, *Otolemur crassicaudatus*, *Perodicticus potto*, and *Varecia variegatus*; and the four "apes" are *Homo sapiens*, *Gorilla gorilla*, *Pan troglodytes*, and *Pongo pygmaeus*

A further observation is that two very distantly related taxa, here cats and prosimians, with a last common ancestor living about 100 million years ago (Bininda-Emonds et al. 2007; Murphy et al. 2007), can occupy the same position in the sensory space. The reason must be a convergence created by similar evolutionary pressures. It should be pointed out, however, that a fourth sensory dimension, i.e., whiskers, discriminates between prosimians and cats. In relation to prosimians, and independent of body size, felids and other carnivores have a better developed system of whiskers and other tactile nose receptors (Muchlinski 2010, Fig. 3).

Sensory organ residuals and ecological variables

Following Nowak (1999), we have classified all 119 species according to diet, and typical habitat; aerial, aquatic, arboreal, fossorial, or terrestrial. A statistical analysis shows that eye and nose sizes are statistically significant covariates of habitat and diet, respectively. First, we shall consider the effect of habitat.

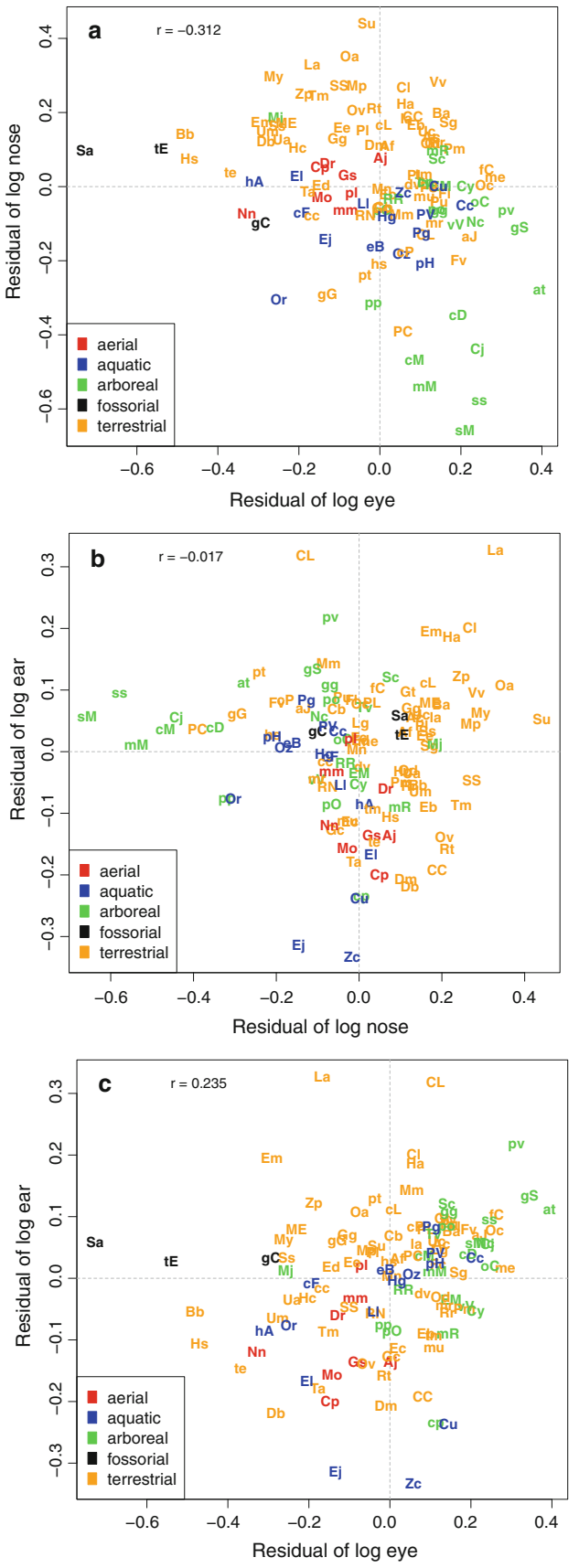


Fig. 5 Two-dimensional projections of the sensory space, plotting ▶ residuals of **a** log nose against log eye, **b** log ear against log nose, and **c** log ear against log eye. Each species ($N = 119$) is identified by a two-letter code (see Table 1 and Supplementary Material Table S2), and *color* coded according to its main habitat (see Figure *inset*). Note the separation between arboreal and terrestrial species

The 119 species are now positioned in Fig. 5a, b, and c in irregular swarms, as order due to body size has been abolished. This differs from the situation before filtering out the scale effect: in Fig. 2 all swarms of data points (including those describing the relation between eye and nose, eye and ear, and nose and ear, respectively) are elongated, as the body size component remains present, even when not made explicit.

By marking the five typical habitats with five different colors, we can see interesting patterns. Plotting log nose residuals as function of log eye residuals (Fig. 5a) produces a clear separation of terrestrial and arboreal mammals, with the terrestrial distributed toward the upper left (large noses and relatively modest eyes), and the arboreal toward the lower right (large eyes and relatively small noses).

Each data point in Fig. 5 is marked by a two-letter code identifying the species as indicated in Table 1 and Supplementary Material Table S2. As expected, the three fossorial species (black) have small eyes. Aerial and aquatic species are both concentrated in the center of Fig. 5a. Among the arboreal species (green), "Mj", the Malayan pangolin *Manis javanica*, is a clear outlier at the upper left. It is an ant-eating arboreal species with small eyes and an impressive nose, a rather peculiar animal. Four terrestrial species are found close together in the lower center, *Homo sapiens* (hs), *Pan troglodytes* (pt), *Gorilla gorilla* (gG) and *Papio cynocephalus* (PC), i.e., relatively large, terrestrial or semi-terrestrial primates. The fourteen remaining primates are all arboreal, and found in the lower right quarter of Fig. 5a. Despite some diverging observations, the data plotted in Fig. 5a support a trade-off between olfaction and vision.

Plotting ears against noses (Fig. 5b) produces a similar separation between terrestrial and arboreal species, i.e., eyes and ears behave in this respect in a similar way in relation to the nose. However, plotting ears against eyes (Fig. 5c) shows no clear separation between terrestrial and arboreal animals, but rather gives an indication of co-operation between these two sensory systems. In the upper right quadrant of Fig. 5c we see a concentration of carnivorous mammals with large eyes and ears, good vision and hearing. Hence, Fig. 5a–c indicate a trade-off between, on one hand vision plus hearing, and, on the other hand, olfaction.

Arboreal mammals separate from terrestrial ones by having larger eyes and relatively small olfactory organs.

Table 1 Species codes in alphabetical order and the corresponding species names with the typical habitats and diets

Code	*Species* (habitat, diet)	Code	*Species* (habitat, diet)	Code	*Species* (habitat, diet)	Code	*Species* (habitat, diet)
Ac	*Antilope cervicapra* (T, H)	Ed	*Erethizon dorsatum* (T, H)	Mj	*Manis javanica* (Ar, I)	pt	*Pan troglodytes* (T, H)
Af	*Ailurus fulgens* (T, H)	Ee	*Erinaceus europaeus* (T, I)	Mm	*Marmota marmot* (T, H)	Pu	*Panthera uncia* (T, C)
Aj	*Artibeus jamaicensis* (Ae, H)	Ej	*Eumetopias jubatus* (Aq, P)	mM	*Macaca mulatta* (Ar, H)	PV	*Phoca vitulina* (Aq, P)
aJ	*Acinonyx jubatus* (T, C)	El	*Enhydra lutris* (Aq, Other)	mm	*Myotis myotis* (Ae, I)	pv	*Pteromys volans* (Ar, H)
at	*Aotus trivirgatus* (Ar, H)	EM	*Eulemur macaco* (Ar, H)	Mn	*Mustela nivalis* (T, C)	RR	*Rattus rattus* (Ar, Om)
Ba	*Bassariscus astutus* (T, Om)	Em	*Elephas maximus* (T, H)	Mo	*Molossus molossus* (Ae, I)	RN	*Rattus norvegicus* (T, Om)
Bb	*Blarina brevicauda* (T, I)	Fc	*Puma concolor* (T, C)	Mp	*Mustela putorius* (T, C)	Rr	*Rupicapra rupicapra* (T, H)
Cb	*Camelus bactrianus* (T, H)	fC	*Felis catus* (T, C)	mR	*Marmosa robinsoni* (Ar, I)	Rt	*Rangifer tarandus* (T, H)
CC	*Capreolus* capreolus (T, H)	Fl	*Lynx lynx* (T, C)	mr	*Macropus rufus* (T, H)	Sa	*Scalopus aquaticus* (F, I)
Cc	*Cystophora cristata* (Aq, P)	Fv	*Prionailurus viverrinus* (T, C)	mu	*Mus musculus* (T, Om)	Sc	*Sciurus carolinensis* (Ar, H)
cc	*Cricetus cricetus* (T, H)	Gc	*Giraffa camelopardalis* (T, H)	My	*Myrmecophaga tridactyla* (T, I)	Sg	*Sylvicapra grimmia* (T, H)
cD	*Cercopithecus diana* (Ar, H)	gC	*Georychus capensis* (F, H)	Nc	*Nycticebus coucang* (Ar, Om)	sM	*Saguinus midas* (Ar, Om)
cF	*Castor fiber* (Aq, H)	Gg	*Gulo gulo* (T, C)	Nn	*Nyctalus noctula* (Ae, I)	SS	*Sus scrofa* (T, Om)
Cj	*Callithrix jacchus* (Ar, Om)	gG	*Gorilla gorilla* (T, H)	Oa	*Orycteropus afer* (T, I)	Ss	*Setifer setosus* (T, I)
CL	*Chinchilla lanigera* (T, H)	gg	*Glis glis* (Ar, H)	Oc	*Oryctolagus cuniculus* (T, H)	ss	*Saimiri sciureus* (Ar, H)
Cl	*Canis latrans* (T, C)	Gs	*Glossophaga soricina* (Ae, H)	oC	*Otolemur crassicaudatus* (Ar, I)	Su	*Suncus murinus* (T, I)
cL	*Canis lupus* (T, C)	gS	*Galago senegalensis* (Ar, I)	Od	*Ovis dalli* (T, H)	Ta	*Tachyglossus aculeatus* (T, I)
cM	*Cercopithecus mitis* (Ar, H)	Gt	*Eudorcas thomsonii* (T, H)	Or	*Odobenus rosmarus* (Aq, Other)	tE	*Talpa europaea* (F, I)
Cp	*Carollia perspicillata* (Ae, H)	Ha	*Hemiechinus auritus* (T, I)	Ov	*Odocoileus virginianus* (T, H)	te	*Tenrec ecaudatus* (T, I)
cP	*Cavia porcellus* (T, H)	hA	*Hippopotamus amphibius* (Aq, H)	Oz	*Ondatra zibethicus* (Aq, H)	Tm	*Tolypeutes matacus* (T, I)
cp	*Caluromys philander* (Ar, Om)	Hc	*Hystrix cristata* (T, H)	PC	*Papio cynocephalus* (T, Om)	tm	*Tupaia minor* (T, I)
Cu	*Callorhinus ursinus* (Aq, P)	Hg	*Halichoerus grypus* (Aq, P)	Pg	*Pagophilus groenlandicus* (Aq, P)	Tv	*Trichosurus vulpecula* (Ar, H)
Cy	*Cyclopes didactylus* (Ar, I)	Hs	*Hemicentetes semispinosus* (T, I)	pH	*Pusa hispida* (Aq, P)	Ua	*Ursus arctos* (T, Om)
Db	*Diceros bicornis* (T, H)	hs	*Homo sapiens* (T, Om)	PL	*Panthera leo* (T, C)	Uc	*Urocyon cinereoargenteus* (T, C)
Dm	*Didelphis marsupialis* (T, Om)	Ia	*Ichneumia albicauda* (T, C)	Pl	*Procyon lotor* (T, Om)	Um	*Ursus maritimus* (T, C)
Dr	*Desmodus rotundus* (Ae, Other)	La	*Loxodonta africana* (T, H)	pl	*Plecotus auritus* (Ae, I)	Vv	*Vulpes vulpes* (T, C)
dv	*Dasyurus viverrinus* (T, C)	Lg	*Lama guanicoe* (T, H)	Pm	*Peromyscus maniculatus* (T, H)	vV	*Varecia variegata* (Ar, H)
Eb	*Equus burchelli* (T, H)	Ll	*Lutra lutra* (Aq, P)	pO	*Phalanger orientalis* (Ar, H)	Zc	*Zalophus californianus* (Aq, P)
eB	*Erignathus barbatus* (Aq, P)	ME	*Meles meles* (T, Om)	po	*Perodicticus potto* (Ar, Om)	Zp	*Zaedyus pichiy* (T, I)
Ec	*Equus caballus* (T, H)	me	*Macropus eugenii* (T, H)	pp	*Pongo pygmaeus* (Ar, H)		

Code species code. Habitats: *Ae* aerial, *Aq* aquatic, *Ar* arboreal, *F* fossorial, *T* terrestrial. Diets: *C* carnivore, *P* piscivore, *I* insectivore, *H* herbivore, *Om* omnivore, *Other* other diet. Data on habitat and diet from Nowak (1999). See also Supplementary Material Table S2

One may thus ask how for example New World monkeys (Strepsirrhines), manage with reduced olfaction. However, small olfactory organs only indicate that an animal is not an olfactory specialist (Pihlström et al. 2005), but may well have high sensitivity to particularly relevant odors. For haplorrhine monkeys in general, such relevant odors have been shown to be, e.g., the scents of fruit alcohols (Laska and Seibt 2002).

Figure 5a indicates that aquatic carnivores on an average have smaller noses and are less olfactory oriented than their terrestrial cousins. All ten seals (pinnipeds) and the two otters, *Lutra lutra* (Ll) and *Enhydra lutris* (El), show negative nose values on the *y* axis. The same applies to "the fishing cat", *Felis viverrina* (Fv). This Asian cat, now named *Prionailurus viverrina*, was classified as terrestrial by Nowak (1999), but is described as a semiaquatic piscivore by Haque and Vijayan (1993). These aquatic and semiaquatic species differ from the 20 terrestrial carnivores of which only seven show negative nose values on the *y* axis.

Figure 6 reproduces the eye–nose relation plot of Fig. 5a, except that the data points (species) are here color coded according to their predominant diet (cf. Supplementary Material Table S2). The six diet categories, from carnivorous to omnivores and "others", are presented in the Figure inset. As for sorting by habitat in Fig. 5a, b, sorting by diet does not produce a random pattern, but a certain degree of group coherence. Carnivorous mammals fall mainly in the upper right quadrant, having medium-sized or large eyes and olfactory organs. Bordering and slightly overlapping with the carnivorous species in Fig. 6 we find piscivores, i.e., fish-eating mammals. On average, they have smaller nose residuals. Interestingly, the "carnivore" with the smallest nose residual (Fv) is the fishing cat *Prionailurus viverrinus* which in fact is a piscivore (see above). A number of previous studies have pointed out that aquatic and semiaquatic mammals tend to have reduced olfactory organs compared with terrestrial mammals (Radinsky 1968; Sánchez-Villagra and Asher 2002; Pihlström 2008).

The herbivores, which form a diverse group composed of grass eaters, leaf eaters, fruit eaters, etc., have a wide distribution all over our plot. The same applies to omnivores. Not surprisingly, the sensory profiles of a number of herbivores and omnivores overlap with those of the carnivorous mammals; the reindeer (*Rangifer tarandus*, Rt) is close to the wolf (*Canis lupus*, cL), the zebra (*Equus burchelli*, Eb) close to the lion (*Panthera leo*, PL), and the mice (*Peromyscus maniculatus*, Pm, and *Mus musculus*, mu) close to the cat (*Felis catus*, fC). Do we witness an "arms race" between prey and predator? The material is too limited to allow a conclusion. The fox (*Vulpes vulpes*, Vv) and the rabbit (*Oryctolagus cuniculus*, Oc) lie within

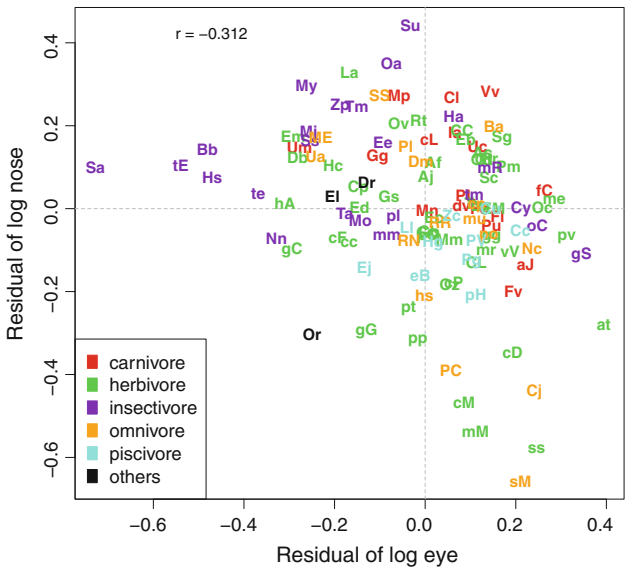


Fig. 6 Two-dimensional projection plotting residual log nose against residual log eye. Here, the species are *color* coded according to their main diet (see Figure *inset*)

the same general region, i.e., the upper right area of this eye–nose plot. However, in terms of residual size the olfactory organ of the fox clearly exceeds that of the rabbit, whereas the rabbit has larger eyes.

The purple-marked insectivores also represent an ecologically and systematically diverse group. In the center, we see a few insect-catching bats, and along the upper border a row of animals with large residual olfactory organs. Most of them are medium-sized or small mammals collecting small invertebrates close to the ground. Two are fossorial.

Whiskers and tactile nose receptors

Whiskers and other types of tactile nose receptors are very important, for instance to shrews, rodents, carnivores and ungulates (Dehnhardt and Mauck 2008; Muchlinski 2010). We have not carried out measurements on tactile organs, but we find it very rewarding to compare Muchlinski′s data with our eye, nose and ear data, and thus add a fourth dimension to our analysis.

Mammalian vibrissae and other mechanoreceptors in the maxillary region (nose region above the mouth) contact the brain via the trigeminal system, i.e., the afferent nerve fibers passing through the infraorbital foramen (IOF). Muchlinski (2010) has noted that the vibrissa count and IOF area are significantly positively correlated. She has determined the IOF areas of 527 mammalian species, 67 of which are also included in our eye database, and 72 in our nose database (Supplementary Material Table S3). Based on these species, we find that the correlation between log eye diameter and log square root of IOF area is weak

($r^2 = 0.292$, $n = 67$). Thus, there is no sign of a specific co-operation of eyes and tactile nose receptors. However, there is significant correlation between log square root of cribriform plate area and log square root of IOF area($r^2 = 0.701$, $n = 72$). Thus, we suggest that nose mechanoreceptors tend to co-operate with olfaction but not with vision. Figure 7 a and b show log eye diameter and log square root of cribriform plate area plotted as functions of log square root of IOF area, together with the least-square regression lines.

Discussion

The evidence for a correlation between sensory organ size and functional efficiency is presented in the "Introduction". There we also note that sensory organs are metabolically expensive, and that an investment in a large organ is an indirect indication of its functional importance.

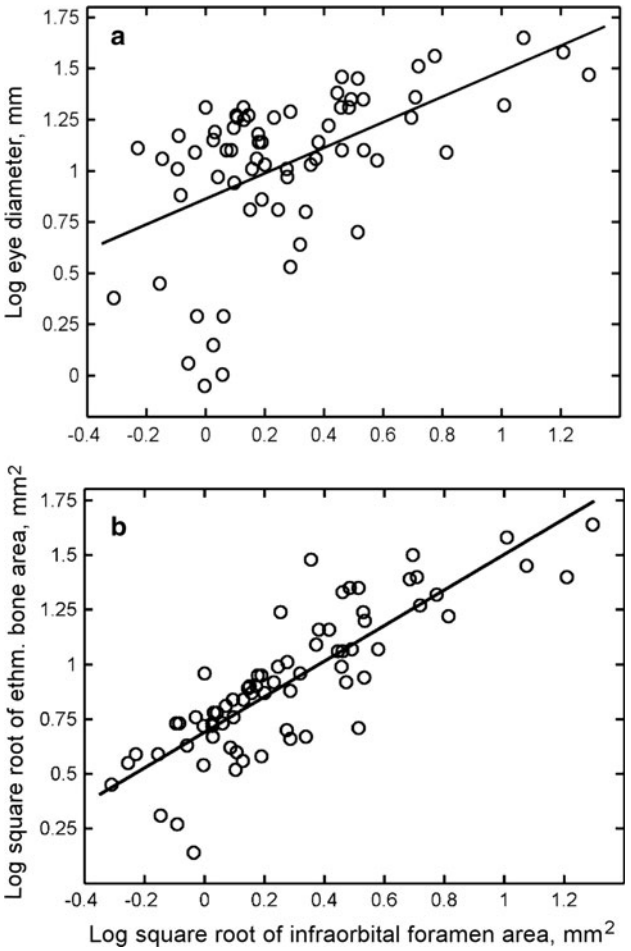


Fig. 7 Log–log plot of **a** eye axial diameter (mm), and **b** square root of cribriform plate area (CP, mm^2) against the square root of the infraorbital foramen area (IOF, mm^2). The lines are least-square regression lines $y = kx + c$. The coefficients of the regression fits are **a** $r^2 = 0.292$ ($k = 0.624$; $c = 0.862$); **b** $r^2 = 0.701$ ($k = 0.812$; $c = 0.689$). Numerical values of the underlying data are presented in Supplementary Material Table S3

Large sensory organs are metabolically costly

Sensory organs are expensive, both during their daily use and as a significant energy investment during ontogenetic development. All neural tissues, including sensory receptor cells and neuronal ganglia, require high ATP consumption. These energy costs may in fact be equally high at rest and whilst processing information, suggesting that animals can never "turn off" their senses (Hodgkin 1964, pp 71–85; Niven and Laughlin 2008; Okawa et al. 2008). The size of each sensory organ is thus under two opposite evolutionary pressures: the need of energy saving favors decreased size, while the need of maximum sensitivity and resolution favors large organs.

The position of a species in the sensory space is then the result of complex evolutionary pressures. The size of a particular organ may be determined by the above balance between energy saving and performance, while the size relations among all three sensory organs are determined by their mutual trade-off or co-operation. Although the evolutionary background is complex, the end result is fairly simple. A large size of a given sensory organ indicates that high performance of the sense in question is important for the survival and reproduction of the species.

How to use residual sizes?

Residual sizes inform us about the size of a sensory organ in relation to the typical size for mammals having a similar body mass. Further, they enable us to relate organs of small animals to those of large ones, thus integrating sensory systems across many scales. However, these latter data must be judiciously interpreted. As pointed out in the "Introduction", sensitivity and resolution may be determined by *absolute* sizes which are hidden by filtering out scale. Thus, when a small and a large species have eyes of similar residual size, it does not mean that their eyes are functionally fully equivalent. However, species with similar body mass, such as lions and zebras, or foxes and hares, may be safely compared in a meaningful way.

Sensory space in an evolutionary perspective

Charles Darwin's (1859) theory of biological evolution by natural selection provided the necessary conceptual framework for integrating the study of sensory organ structure and function with behavioral ecology. Early examples of such an approach are Karl von Frisch's studies on animal color vision and bee orientation. Recently, the

coming together of computational paleobiology and proxy-based research on long-term climate change has led to the development of a quantitative synthetic model called "ecometrics" (Eronen et al. 2010; Polly et al. 2011; Liu et al. 2012). Ecometrics focuses on easily measurable traits, whose structure is closely related to their function, and whose function interacts directly with the organism's local environment. The approach of the present study can be seen as a special case of ecometrics and could, if applied to fossil species, reveal behavioral information that is both valuable (e.g., Liow et al. 2008) and difficult to obtain by other means.

Multisensory signaling and analysis

Sensory organ co-operation based on combined visual, olfactory and/or auditory cues is widely spread among mammals (Heffner and Heffner 1992; Bradbury and Vehrencamp 1998; Seyfarth and Cheney 2009). Considering the three senses and 119 species studied here, there seems to be a preferred co-operation between vision and hearing. That inference is based on the observation that ear size is a clearly significant covariate of eye size, while olfactory organ size and eye size are more independent (Fig. 2). Many recent studies have explicitly supported a close co-operation of vision and hearing. The simultaneity of signals from both vision and audition is useful for telling a human observer that she is dealing with one particular source and not just with a random mixture of visual and auditory noise (Burr et al. 2009), and both horses and humans are known to combine vision and hearing in their recognition of individuals within their own herd or social group (Proops et al. 2009). Apparently the interaction between human vision and hearing is related to neural connections between neighboring visual and auditory areas in the central nervous system (King 1999; Sams et al. 1991; Musacchia et al. 2006). Co-operation between the two senses is likely to have been essential for the evolution of human speech (e.g. Chandrasekaran et al., 2011).

Olfaction and tactile senses

Water shrews foraging underwater seem to use a combination of olfaction and tactile detection (using vibrissae), but not eyesight, for prey capture (Catania et al. 2008). Possibly, large mammals tend to combine vision and hearing, while smaller ones, like shrews and small rodents, may combine olfaction and tactile senses, as also suggested by data presented in Fig. 7. It is well known that small mammals such as shrews have well-developed vibrissae, but poor vision as a result of the inherent optical limitations to very small eyes (Suthers and Wallis 1970; Catania 2005 and references therein).

In this study, we have estimated the relative importance of three mammalian sensory systems by comparing the sizes of three organs. With a similar aim in mind, Sánchez-Villagra and Asher (2002) measured the area of the infraorbital foramen through which afferent nerves from the vibrissae pass toward the brain; thus they could estimate the dimension of this branch of the trigeminal system. As described above, Muchlinski (2010) investigated the quantitative relation between vibrissae and infraorbital foramen area. In addition, Sánchez-Villagra and Asher (2002) measured the width of the anterior cranial fossa to estimate the extension of the contact between olfactory structures and the brain.

It is encouraging to note that our independent studies have reached similar conclusions; in comparison with related terrestrial taxa, a wide range of semiaquatic piscivore mammals were observed to have a less-developed olfactory system (Figs. 5a, 6; Sánchez-Villagra and Asher 2002), and in many small-sized mammals we note a correlation between well-developed olfactory organs and whiskers (Fig. 7b being based on Muchlinski 2010).

When comparing olfaction in fish and aquatic amphibians with olfaction in terrestrial mammals, we realize that the mechanisms differ in many levels, both regarding anatomical and histological structures (Reiss and Eisthen 2008; Pihlström 2008), and regarding the cellular and molecular processes that bring the water-soluble or volatile odorants in contact with the olfactory receptor cells (Hemilä and Reuter 2008; Eisthen and Schwenk 2008). Many aquatic and semi-aquatic mammals descend from terrestrial forms, and hence we may assume that the reduced size of the olfactory organs of many semi-aquatic mammals may reflect inertia in evolutionary processes transforming a terrestrial system into an aquatic one.

Is olfaction an evolutionarily primitive sense?

A recurrent argument in evolutionary sensory biology is the idea of the sense of smell as a "primitive" sense that has been increasingly reduced during evolution, in parallel with more refined visual systems. This idea may have some justification in the case of haplorrhine primates, i.e., monkeys and apes, being visually oriented mammals with small cribriform plates and reduced repertoires of functional olfactory genes (Rouquier et al. 2000; Smith et al. 2007). However, some more recently derived taxa, such as extant carnivores and artiodactyls, seem to have sensitive olfaction and well-developed olfactory structures (Pihlström 2008; Pihlström et al. 2005). A wide range of semiaquatic mammals have been observed to have an underdeveloped olfactory system relative to terrestrial taxa, as a convergent specialization to their aquatic environment

(Sánchez-Villagra and Asher 2002). Taken together, there is no *general* reduction of the sense of smell in mammals.

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